

# Inequality in Resilience: Understanding Household Electricity Consumption During Load Shedding

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## Abstract

Households with sufficient financial means are increasingly adopting solar home systems (SHS) in Cape Town, partly to mitigate experienced and anticipated future load shedding. These capital-intensive solutions are out of reach for many lower-income households. In this paper, we investigate whether SHS adoption significantly affects household electricity consumption during load shedding, to examine the extent to which SHS enhance household resilience to unreliable power. We first build a deep learning model using the Mask2Former (M2F) infrastructure to identify the locations and capacities of SHS within the city. We then merge SHS location data with detailed information on household electricity transactions and load shedding over time. Our fixed-effects econometric model results indicate that SHS reduce electricity consumption from the grid by approximately 20% on average, and even more during load shedding. This effect dissipates over time, however, and is likely a lower bound due to uncertainty in our SHS data. In contrast, households without SHS experience minimal to no reduction in consumption even as outages increase, which suggests that their lost consumption is largely shifted in time. These results show that SHS are effective in reducing the need for costly behavioral adjustments under load shedding, for those who can afford them.

## Introduction

Globally, many low- and middle-income economies are facing the dual challenge of expanding high-quality electricity service and fostering an energy transition to more climate-friendly fuel sources. The latter presents considerable opportunities, as costs of solar (National Renewable Energy Laboratory (NREL), 2021) and wind (Energy Markets & Policy, Berkeley Lab, 2021) have declined dramatically in recent decades. Yet there are also considerable risks, due to resource intermittency and the many issues that arise when integrating distributed generation sources into conventional grid systems. South Africa, as a developing economy with a heavily coal-dependent electricity sector – 80% of generation comes from coal-fired power plants (IEA, 2025b) – has also struggled with reliability problems, and therefore offers a particularly stark example of this dual challenge. The country is responsible for the largest share (about

30%) of carbon dioxide emissions on the African continent (IEA, 2025a), despite accounting for only about 4% of its population (Statistics South Africa, 2024).

Over the past two decades, the electricity system in South Africa has buckled under periodic periods of disruptive load-shedding, due to a complex set of institutional, financial, and technological factors, with major implications for the profitability of firms and for household well-being. Moreover, vast differences exist in how South Africans use and depend on the energy system for their livelihoods across a range of dimensions, including geography, ethnicity and race, socio-economic conditions, and urbanity. As such, the country also demonstrates the challenges of promoting equity in energy transition. Indeed, this was part of the motivation behind the South African government’s recent entering into a Just Transition Partnership Agreement during the 26th Conference of the Parties (COP26) in 2021, which is supported by \$8.5 billion USD of funding from the multi-government International Partners Group (IPG), but requires much more funding to secure South Africa’s energy future. This context makes South Africa a valuable setting to evaluate the relationship between a transitioning energy system, system reliability, and equity.

This paper focuses on the particular context of Cape Town, South Africa’s most unequal city (Esri, n.d.), which is a microcosm of the larger energy transition equity issues facing the country as a whole. To cope with frequent electricity outages during load shedding periods, many households and businesses in Cape Town have invested in backup electricity systems, mostly solar home systems (SHS). These systems provide two key advantages to consumers with the means to cover the high upfront costs of installation. First, they allow these utility customers to reduce their exposure to unreliable national grid power. Second, over the longer term, they may deliver cost savings due to decreased purchases from the grid and the electricity consumption cost offsets provided by Cape Town’s feed-in tariff. Cape Town was the first city in the country to buy excess generation from SHS owners in exchange for cash or credits (City of Cape Town, 2024). And while SHS uptake contributes to Cape Town and South Africa’s renewable energy transition, these systems are currently primarily accessible to upper-income households.

This affordability challenge has resulted in a high degree of clustering of SHS in wealthier neighborhoods, and only limited installation outside of these locations. This, in turn, has led to questions about the equity of the renewable transition, heightening the debate over climate and energy justice in the country. When high-income households become relatively more independent of the grid through solar installations, they receive cost offsets on their utility bills from the feed-in tariff and contribute less to utility revenue. Utilities or distributors, in turn, become more reliant on revenue collected from lower-income households (Baker et al., 2015). This may lead to utilities needing to further raise rates on an ever narrower base of less wealthy customers. In addition, wealthier neighborhoods potentially receive higher investment when municipalities’ work to upgrade their electricity infrastructure to accommodate the influx of SHS (Oliver, 2021), further exacerbating inequality in infrastructure quality and costs. The underlying dynamics of SHS adoption make it clear that distributional impacts and systemic implications need to be better understood.

The nature of solar adoption incentives should also be carefully considered. To date, Cape Town

residents have been incentivized to install grid-connected solar home systems using several instruments. For example, the 2024/2025 budget included a residential feed-in tariff of \$0.07/kWh (inclusive of VAT), with an additional incentive of \$0.02/kWh (inclusive of VAT) applicable until January 2026 (City of Cape Town, 2025b). Non-residential consumers also receive this additional incentive, on top of a base feed-in tariff ranging from \$0.05 to \$0.06/kWh (inclusive of VAT). In addition, for residential solar panels purchased between March 2023 and February 2024, buyers were eligible to claim a 25% rebate (up to \$900) on their taxes. All of these incentives largely favor wealthier consumers, who either pay more taxes or can afford the upfront costs of solar installation and benefit from the cost offset of the feed-in tariff. To close revenue gaps, lower-income households, meanwhile, may be subjected to increased residential tariffs (Baker et al., 2015).

Incentivizing households to adopt SHS is a conflict of interest for the national utility and electricity distributors, as household self-production decreases both utility and local municipality revenue (McKay and Hendricks, 2022). Cape Town relies on revenue from electricity distribution, and as of now, there is no mechanism for SHS users to support the grid, with potentially harmful implications for existing infrastructure maintenance and public finances (Baker and Phillips, 2019). This may decrease the government’s ability to provide affordable services to low-income residents. For example, a 2015 study found that when high-income households buy SHS, money is taken away from programs such as Free Basic Electricity. A fixed connection fee, which would apply to both grid and PV users, could help alleviate this, making the PV transition potentially revenue neutral, but pricing designs must be considered holistically (Mayr et al., 2015b). Indeed, this tension between utility and municipality revenue and SHS adoption poses another systemic barrier to achieving higher adoption rates.

The study makes several contributions related to the understanding of Cape Town’s potential energy transition. First, most published studies on this subject date from the early 2000s, when South Africa had a substantially different generation portfolio and energy demand (Statistics South Africa, 2012), and when solar generation was largely irrelevant. Since that time, considerable work has delved into the impact of decentralized solar in South Africa, beginning with a review of the early SHS concession program (Wlokas, 2010), and other studies focused on rural consumers (Azimoh et al., 2015; Stojanovski et al., 2017). Relevant to Cape Town, older work estimated that households with grid-connected solar home systems in Cape Town could feasibly offset 50% of their electricity needs (Chidi et al., 2012), but that adoption would be concentrated among high-income households and lead to reduced revenue for the utility (Mayr et al., 2015b) – an implication that has not been adequately studied recently. More recent studies have estimated the benefits to consumers of grid-connected PV (Marais et al., 2019; Senatla et al., 2020), without examining system implications. There are few studies that integrate technical, economic, and equity dimensions in an urban context with persistent load-shedding, such as Cape Town.

Second, this work speaks to an urgent policy need. Though the load shedding in South Africa has somewhat abated since its peak in 2023-2024, the planned phase-out of aging generation assets over the coming decades, and the lack of a detailed understanding of the diffusion of decentralized solar generation, raise the risk that electricity supply and demand imbalances may soon recur. Recent work has demonstrated the high costs of load shedding in the country (Naidoo, 2023; Bhorat and Köhler,

2024), and how it varies substantially across industries and sectors (Goldberg, 2015; Walsh et al., 2020). Accordingly, a major goal in Cape Town’s 2050 energy strategy, published in 2023, is to alleviate higher load shedding stages when they are imposed by the national utility (City of Cape Town, 2019).

In addition to providing updated research on Cape Town’s energy transition and addressing an urgent policy need, this study also seeks to better understand how SHS adoption affects grid energy consumption and payments. As adoption increases, it becomes more and more important to understand consumption behavior for households who own such systems, to manage load and forecast demand. SHS adoption may lead households to smooth their consumption during periods of load shedding, reducing strain on the grid, while households without backup systems shift consumption to before or after outage periods. The complex behavioral adjustments have important implications for the equity of the energy transition in Cape Town, as low-income households bear the brunt of outage burdens, while households with SHS are protected. On the other hand, if SHS households can smooth consumption during load shedding, SHS adoption may in fact be beneficial for the grid, as load shifting around outage periods may increase grid strain. Understanding electricity consumption behavior for households both with and without these systems also helps to clarify potential revenue impacts for both the municipality and the national utility, and can inform better tariff design.

## Background & Context

### Relevant Literature

Research on grid-connected PV systems in Africa remains limited, particularly in contexts with load shedding (Brunet et al., 2018) and for household consumption behavior in South Africa (Mutumbi et al., 2022). A few papers have examined factors influencing behavior during blackouts, finding that outage duration, income (Abbou et al., 2022; Igel, 2024), and age of household occupants (Dulam et al., 2025) are influential. A Cape Town study suggests that outages lead to routine changes where priority is given to essential appliances (Igel, 2024). One study on South Africa finds that SHS have the potential to improve the reliability of the national grid, benefiting households with and without SHS (Mabunda, 2021).

On the other hand, several studies have identified unintended consequences of battery charging during load shedding. For example, although 25% of households using batteries could mitigate 90% of load shedding (Ritchie et al., 2023), it also increases grid stress. 15% of South African households charge external batteries on the grid to prepare for load shedding, increasing the national load by twice the amount shed (Ritchie et al., 2024). This adverse effect has also been documented in Pakistan, where Zubair et al., 2021 find that battery-charging consumers increase the load on the national grid by a third, requiring the grid to shed about that same amount of load. In fact, the authors argue that if consumers stopped charging their batteries from the grid, the country may not need to implement load shedding. To separate the impact of battery charging on the national grid load, authors have emphasized the importance of using PV for charging (Ritchie et al., 2024; Zubair et al., 2021), leading to one of the



more effective ways to mitigate load shedding (Twala et al., 2023). These studies suggest that, without appropriate management, charging batteries from the grid can exacerbate load shedding, a dynamic highly relevant to Cape Town.

Various studies assess the financial feasibility of solar home systems, noting that low-income households are not incentivized to invest due to the high fixed costs (Senatla et al., 2020) and non-incentivizing feed-in tariff (van Schalkwyk and Scholtz, 2024), in addition to low average consumption (Senatla et al., 2020). Not only are the upfront costs unattainable, but South Africa has not effectively subsidized the technology or provided affordable loans (McKay and Hendricks, 2022) even though cost savings have been found to be the most influential way to get households to adopt (van Schalkwyk and Scholtz, 2024). To surpass this obstacle, other countries, such as Uganda and Kenya, have used public/private partnerships to provide access to loans for installing SHS (Damo et al., 2024). Further research needs to be conducted in Cape Town to understand how policy interventions can improve adoption rates. Most high-income households were projected to adopt SHS by the mid-2020s by Mayr et al., 2015a, which, if the case, will lead to a decrease in municipality and utility revenue, and will be further exacerbated by non-compliance (Dippenaar and Bekker, 2025).

## Electricity Context in Cape Town

Cape Town provides a unique case to study grid-connected PV adoption due to its stratified uptake and frequent load shedding. Electricity in Cape Town is predominantly supplied by the national utility, Eskom, which also distributes about 25% of the electricity in the city, leaving the remaining 75% distributed by the municipality through its own distribution networks (Cape Town Green Map, 2025). Load shedding has been a nationwide issue since 2007 (Bank, 2025) and occurs when there is a supply shortage. In Cape Town, load shedding is classified into eight stages, with one being the least severe. In stage one, 12 of the 16 areas in Cape Town endure two and a half hours of load shedding per day, with one area affected at a time. In stage eight, each two-and-a-half-period cuts off power for eight areas, with a different set of eight for each two-hour period, meaning the same area will experience a two and a half hour power cut multiple times a day. These “areas” are also known as “blocks” with blocks 1 - 16 supplied by the City and 17-23 supplied by Eskom (City of Cape Town, 2025c).

These more recent periods follow three distinct periods of consistent blackouts from 2007-2009, 2013-2015, and 2018-2019, which have been estimated to cost somewhere between \$2 (Walsh et al., 2020) and 2.5 (OECD, 2025) billion USD. This amount, which excludes long-term impact on growth, such as loss in investments or other non-GDP effects, is approximately equal to the 2008/2009 financial crisis. The impact of load shedding from 2020 to Q1 of 2023 is approximated to account for about 3% of GDP in 2023 at \$13 billion USD (Kay et al., 2023).

According to data supplied by the City of Cape Town, the municipality experienced about 150 days of load shedding in 2022, 300 days in 2023, and 70 days in 2024.

The costs, financial and social, of load shedding in South Africa have been well documented (Consulting, 2025; Libera, 2025b), and include decreased safety due to lack of lighting, less time teaching in schools, particularly for poorer communities, and lack of power in health centers that can lead to a

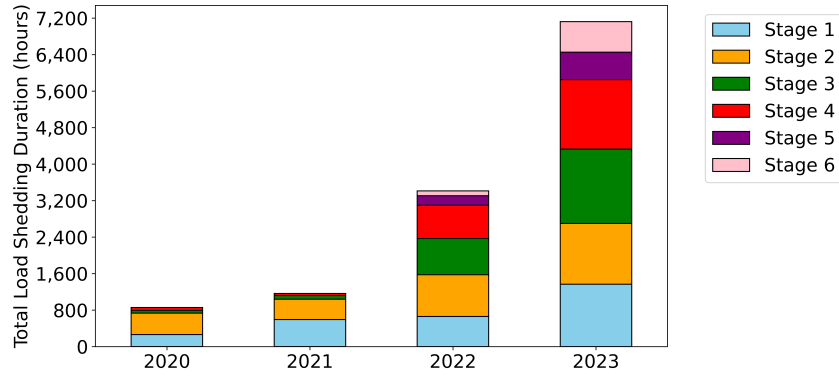


Figure 1: Load shedding by stage from 2020 to 2023. Data provided by the City of Cape Town.

multitude of risks, including compromising medicinal storage, or unavailability of machinery and treatments for patients. A 2025 report by FTI Consulting attributes part of the reduction in load shedding in 2024 to increased SHS capacity, totaling over 6 GW, exceeding the stage 6 load shedding. Although load shedding has been less frequent, scholars have deemed this trend unsustainable due to Eskom’s increased reliance on diesel to run open cycle gas turbines (OCGT) plants (Consulting, 2025), which has led the utility to apply for a 40% average tariff increase across consumer categories for 2025/2026 (Libera, 2025b). This follows a decade of increasing prices (Igel, 2024). In addition, the Eskom Debt Relief Act of 2023 has provided Eskom with over 9 billion USD as of 2025 (CNBC Africa, 2025). Others have suggested that South Africa still lacks enough generating capacity (Jacobs, 2025).

Although Cape Town is mostly supplied by Eskom-owned power plants, the city owns and operates the 180 MW Steenbras hydro pumped storage, a 7 MW PV plant (Atlantis), a 42 MW gas turbine (Roggebaai), and a 36 MW diesel gas turbine (Atholone) (OpenInfraMap, 2025). The Steenbras plant is used mainly to mitigate against load shedding, where it can mitigate up to two stages, where one stage in the summer is 40 MW and 60 MW in the winter (Nassiep, 2023). The maximum demand for the city, reported in 2023, is 1.7 GW.

## SHS Policy Background in Cape Town

Between 1 March 2023 and 29 February 2024, South African residents could take advantage of a residential solar panel tax rebate of 25% up to approximately \$900 USD towards their Personal Income Tax liability (South African Revenue Service, 2024). Typical residential panels can cost between \$150 – \$300 each (Africa, 2025), with installation costs typically around \$600 – \$900, making residential systems cost between \$4K – \$15K (Procompare, 2025), still a significant barrier.

To install an SHS for your home, the City specifies a few requirements. For example, you must first register your system and get approval before buying and installing your system, a process that can take a few months (Blue Sun Energy, 2025). As of October 2023, the City no longer authorizes standby or off-grid systems and only approves grid-tied and grid-tied hybrid systems. A household that opts not to sell excess electricity back to the grid gets a prepayment meter for free, while feed-in customers have to pay for a bi-directional meter. Feed-in households are allowed to sell more electricity than they

receive and can be paid through the Cash for Power programme. (City of Cape Town, 2025f). A 2025 study finds that approximately two-thirds of SHS systems are not registered with the City, making them illegally connected (Dippenaar and Bekker, 2025). This trend has been attributed to the over \$1,000 cost of compliance, including a bidirectional meter, engineering sign-off, and inverter. The fixed fee for SHS customers decreased in 2020 to encourage consumers to adopt systems, equating the fixed fees for residential consumers with and without SHS, while the feed-in tariff increased (Dippenaar and Bekker, 2025). However, the City has been criticized for penalizing SHS owners by substantially increasing the fixed fee for residential consumers on the Home User tariff (for households with property value greater than \$60,000 USD) by about 40% and reducing the per kWh fee (City of Cape Town, 2025a). Because households with SHS use the national grid as a backup electricity source, they will be most impacted by this change, as they pay substantially more per month with the fixed fee and now less for their lower consumption (Labuschagne, 2026).

Although it never materialized, the National Energy Regulator of South Africa (NERSA) approved a national feed-in-tariff for renewable technology starting in 2009, aimed at achieving 10 TWh of renewable energy by 2013 (Odeku et al., 2011). In 2023, Cape Town became the first municipality to approve a feed-in tariff for rooftop PV (Hill-Lewis, 2023), and the “Cash for Power” program launched in 2024, allowing households to sell excess generation to the grid for credit on the municipality bill (BusinessTech, 2024). As of February 2025, there were about 1,800 households and businesses participating in the program (Libera, 2025a), up from 900 in April 2024 (Democratic Alliance, 2024). Although studies on the impact of the feed-in tariff in Cape Town are yet to materialize, Kelvin and John, 2021 found that in Perth, households with solar home systems plus batteries have become net producers. Because the tariff is flat, these households export power during periods of low demand, which replaces cheaper utility generation, leading the authors to suggest a dynamic feed-in tariff.

## Tariff Structure

Cape Town uses an inclining block tariff structure for electricity prices, which, by design, subsidizes electricity for low-consuming households (Kelly and Geyer, 2018). The tariff a prepaid household is assigned to depends on property value, installed meter type (prepaid in this case), and monthly usage. If a household meets the qualifications of the Lifeline Tariff, including a property value below \$30,000, it receives Free Basic Electricity of 25 kWh with less than 250 kWh monthly consumption or 60 kWh with 250 – 450 kWh/month (City of Cape Town, 2025e). Households with a property value between \$30,000 and \$60,000 are on the Domestic Tariff, and the rest are on the Home User tariff. All postpaid customers and prepaid households with a grid-connected SHS are on the Home User tariff. Tariffs used in this analysis are shown in Table 1.

Table 1: Residential tariff categories and attributes (including VAT). Rates show the range of 2020–2023 values in USD.

| Category  | Property Value | Monthly Use (kWh) | Block 1 (\$/kWh) | Block 2 (\$/kWh)     | Access Fee (\$/month) |
|-----------|----------------|-------------------|------------------|----------------------|-----------------------|
| Lifeline  | <30,000        | <450              | 0.08–0.11        | 0.16–0.22 (350+ kWh) | –                     |
| Domestic  | 30,000–60,000  | >450              | 0.13–0.20        | 0.16–0.25 (600+ kWh) | –                     |
| Home User | >60,000        | –                 | 0.12–0.18        | 0.16–0.25 (600+ kWh) | 10.00–15.00           |

Notes: Home User tariff also includes all residential postpaid customers. “–” indicates not applicable.

## Data

This project uses multiple datasets to construct the variables used in the empirical model. In Figure 2, original datasets, provided by the City of Cape Town or Overture Maps, are outlined in blue. Electricity transactions and aerial imagery are then transformed into monthly observations for the former, and predicted SHS locations and capacities for the latter. Monthly transaction data form the output variable. The predicted SHS locations and capacities are then connected to consumer contracts by matching both with Cape Town building data provided by Overture Maps. This approach allows us to identify all Cape Town households with rooftop solar panel installations, excluding non-generating assets such as solar water heaters. Spatial load shedding data is provided by the City of Cape Town, which we match with each household to determine load shedding duration per month for each consumer.

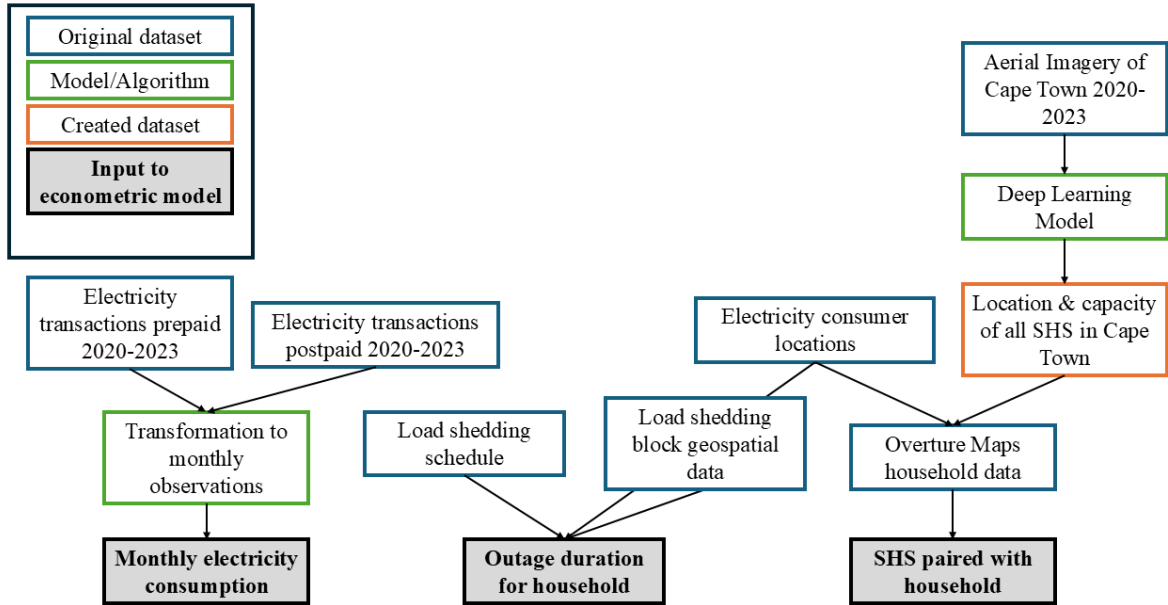


Figure 2: Data map illustrating variable construction for the empirical analysis

## Electricity Consumption and Load Shedding Data

This study utilizes data from the City of Cape Town on electricity transaction payments, load shedding schedules, and geospatial data. The electricity transaction data for both prepaid and postpaid households spans from January 2020 to December 2023, and includes details on each payment, including the amount

of electricity purchased (kWh), applied tariff(s), and any necessary financial adjustments. We only use contracts for which we have location data (XX %). Table 2 shows the number of contracts in each tariff category for each year in our study.

Table 2: Number of households on each tariff over time for prepaid and postpaid consumers. Data provided by the City of Cape Town.

| Year | Prepaid  |          |           | Postpaid  |
|------|----------|----------|-----------|-----------|
|      | LifeLine | Domestic | Home User | Home User |
| 2020 | 74,981   | 89,771   | 110,491   | 55,549    |
| 2021 | 81,882   | 99,242   | 127,454   | 54,453    |
| 2022 | 88,156   | 101,440  | 141,501   | 50,647    |
| 2023 | 91,719   | 107,624  | 169,912   | 46,270    |

Our prepaid datasets consist of newer transaction data from March 2021 to December 2023, where the exact date of the transaction is recorded, and older transaction data, where only the month of the transaction is recorded. Since prepaid electricity can be purchased at any time, the intervals between transactions vary across customers. To construct a monthly panel, we follow the approach of Jack and Smith, 2020 by assuming a constant rate of consumption between payments and that consumers do not accumulate credits on their meter – that is, each consumer waits until their meter has zero credits before making another purchase. This allows us to divide the amount of credits purchased equally among the days before the next purchase is made. We assume that all purchases recorded in the older prepaid dataset are made on the first day of the month. Figure 3 shows the result of transforming prepaid transactions into monthly observations for each tariff category.

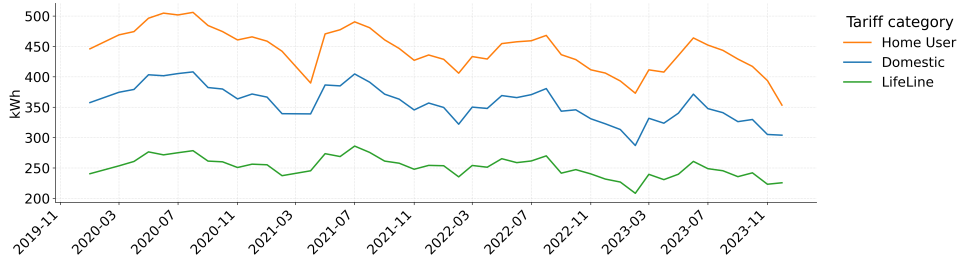


Figure 3: Average monthly consumption for prepaid households in each tariff category from January 2020 to December 2023. Data provided by the City of Cape Town

In addition to prepaid transactions, our analysis also includes postpaid households. We apply the same monthly aggregation used for the prepaid data. Because the postpaid records are in the same format as the old prepaid data, we similarly assume that all purchases occur on the first day of the month. Figure 4 presents the results of this transformation, illustrating that postpaid households, although larger overall consumers, follow the same overall declining consumption trend over the years as the prepaid customers.

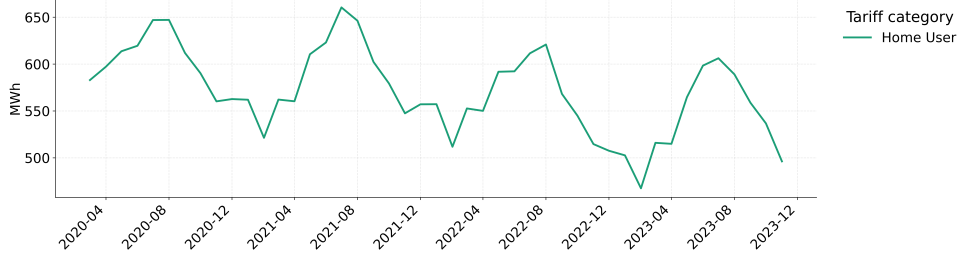


Figure 4: Average monthly consumption for postpaid households in each tariff category from January 2020 to December 2023. Data provided by the City of Cape Town.

It is important to note that we do not see consumption directly from SHSs. The transaction data only provides information on grid power purchases, so any consumption during outages or normal operations from SHS is not considered in this analysis.

In addition to the electricity transaction data, we also use spatially explicit load shedding data for the same time period. Cape Town is divided into 16 areas, or blocks, that are supplied electricity directly by the City rather than the national utility, Eskom. For the rest of the city, load shedding is determined by Eskom and not considered in this analysis.

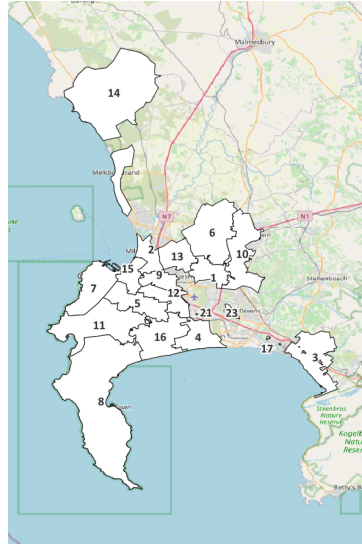


Figure 5: Load shedding blocks in Cape Town (of Cape Town, 2025). Blocks 1-16 are supplied by the municipality.

Load shedding is assigned to each block for about two and a half-hour periods, with the last half hour used for transition. Under Stage 1 load shedding, 12 blocks each experience one outage per day for two and a half hours, with the specific blocks and order varying by day of the month. Stage 2 load shedding assigns two blocks to each load shedding period, and stage 3 assigns three blocks to each period. The last load shedding stage, stage 8, cuts power to eight blocks for each outage period.(City of Cape Town, 2025d)

Over the period of our study, load shedding increased in duration (see Figure 1 for total load shedding over time and Figure 6 for spatial variation across and within years). Cape Town’s load shedding schedule assigns a block or blocks for each two and a half hour period based on the day of the month.

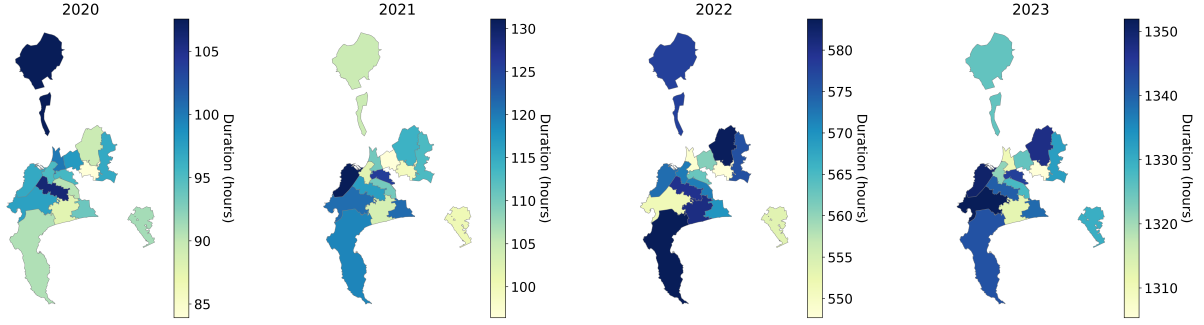


Figure 6: Load shedding duration variation within and across years. Data provided by the City of Cape Town

## SHS Identification from Aerial Imagery

Previous research has found that although Cape Town requires households with SHS to register with the city, only about one third comply to avoid the monthly fee (Dippenaar and Bekker, 2025). As a result, the City’s SHS registration database is incomplete and unsuitable for our analysis. To address this limitation, we built our own deep learning model to identify SHS in aerial imagery. Our model provides the GPS locations and surface areas of each identified solar panel, classifying the systems as either electricity-generating or water-heating to ensure geysers are not mistaken for generation assets.

We use the Masked-attention Mask Transformer (Mask2Former) model (Cheng et al., 2022), which has been used in other deep learning applications such as identifying cancer cells in pathology images (Sheng et al., 2023) and in precision agriculture applications (Darbyshire et al., 2023).

We retain high-resolution aerial imagery from the City of Cape Town from 2020 to 2023. Images are taken in January in most cases, except in 2021, which was taken in March 2020. We assume an installation date for a panel that is first identified to be in June of the previous year. For more information on the model, see Appendix A, and Appendix B for capacity calculation information.

The distribution of SHS capacity for each year is shown in Figure 7. There is an overall increasing trend in the adoption of both small and large systems over time.

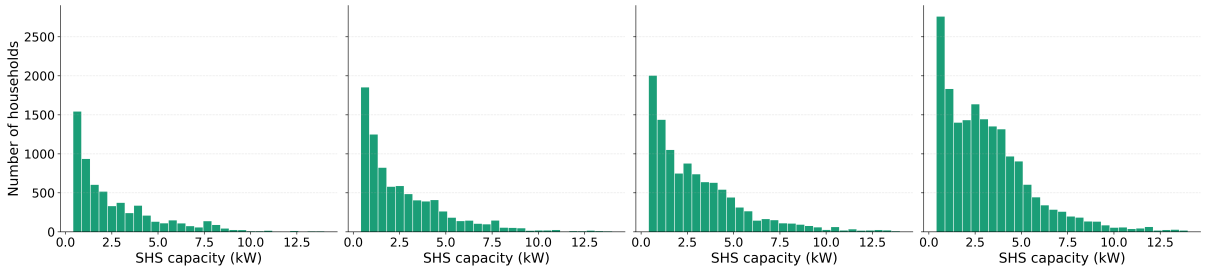


Figure 7: Distribution of SHS capacity in Cape Town from 2020 to 2023

With SHS location data from the Mask2Former (M2F) model, we match each SHS with the nearest household using data from Overture Maps (Overture Maps Foundation, 2025) and, therefore, to the electricity transaction data provided by the City. For each year, we have about 400,000 – 430,000 contracts with monthly consumption data. We have identified SHS for about 1-9% of these households

each year, with median capacity growing each year.

Table 3: PV Contracts and SHS Installations by Contract Type and Year

| Year | Type     | Total Contracts | SHS Households | Median kW |
|------|----------|-----------------|----------------|-----------|
| 2020 | Postpaid | 71,501          | 2,193 (3.1%)   | 2.7       |
|      | Prepaid  | 351,607         | 4,370 (1.2%)   | 1.7       |
| 2021 | Postpaid | 68,726          | 2,799 (4.1%)   | 2.9       |
|      | Prepaid  | 365,479         | 5,941 (1.6%)   | 1.8       |
| 2022 | Postpaid | 64,320          | 3,528 (5.5%)   | 3.4       |
|      | Prepaid  | 343,610         | 8,358 (2.4%)   | 2.3       |
| 2023 | Postpaid | 59,388          | 5,127 (8.6%)   | 3.5       |
|      | Prepaid  | 362,020         | 14,617 (4.0%)  | 2.7       |

## Empirical Framework

### Model 1: Impact of Outages on Consumption

$$y_{it} = \beta_1 \text{OutageDuration}_{it} + \eta_i + \tau_t + \varepsilon_{it}, \quad (1)$$

Equation 3 regresses logged monthly electricity consumption  $y_{it}$  on  $\text{Outage}_{it}$ , the total load shedding hours the household experienced based on their Block ID. We use contract ( $\eta_i$ ) and month/year ( $\tau_t$ ) fixed effects. Standard errors are clustered at the household level. Assuming that  $\varepsilon_{it}$  is uncorrelated with owning an SHS and the capacity of the system, the  $\beta$  and  $\delta$  values should tell us the impact of SHS during load shedding.

Previous work has shown that high-income households are more likely to adopt SHS, mostly due to the high upfront costs (Mutumbi, 2024), making SHS adoption not random. For this reason, we include household fixed effects to control for time-invariant unobserved factors to focus on variation within a household rather than across households. Additionally, we use month and year time fixed effects to account for any unobservable factors that change over time, affecting the entire sample, such as economic trends or the impact of COVID. We separate the month and year into separate fixed effects to detect seasonal trends, such as higher consumption in the winter.

### Model 2: Adoption Effects

To understand how electricity consumption is affected by SHS ownership, we extend the fixed-effects regression by including a binary indicator for SHS adoption,  $\text{SHS}_{it}$ :

$$y_{it} = \beta_1 \text{Outage}_{it} + \beta_2 \text{SHS}_{it} + \beta_3 (\text{Outage}_{it} \times \text{SHS}_{it}) + \eta_i + \tau_t + \varepsilon_{it}, \quad (2)$$

where all other variables are defined as in Model 1. In this specification,  $\beta_1$  captures the effect of an hour of load shedding for households without SHS,  $\beta_2$  measures the percent difference in monthly consumption



between households with and without SHS, and  $\beta_3$  represents the additional change in consumption during an hour of load shedding for households that have installed PV, relative to households without PV.

### Model 3: Capacity Effects

We next examine how the size of PV systems influences consumption behavior by incorporating a continuous SHS capacity variable, PV size<sub>it</sub>, into the baseline regression.

$$y_{it} = \beta_1 \text{Outage}_{it} + \beta_2 \text{PVCapacity}_{it} + \beta_3 (\text{Outage}_{it} \times \text{PVCapacity}_{it}) + \eta_i + \tau_t + \varepsilon_{it}, \quad (3)$$

where all other variables are defined as in Model 1. In this specification,  $\beta_2$  measures the effect of an additional 1 kW of PV capacity on monthly electricity consumption, while  $\beta_3$  indicates the incremental change in consumption during an hour of load shedding with an additional 1 kW of installed PV.

### Model 4: Event Study

We supplement the first three models with an event study to understand the effects of PV ownership over time:

$$y_{it} = \delta \text{Outage}_{it} + \sum_{\tau \neq -1} \beta_\tau \mathbf{1}\{\text{event\_time}_{it} = \tau\} + \sum_{\tau \neq -1} \gamma_\tau \mathbf{1}\{\text{event\_time}_{it} = \tau\} \cdot \text{PVCapacity}_i + \eta_i + \tau_t + \varepsilon_{it}, \quad (4)$$

where  $\beta_\tau$  measures the effect of being  $\tau$  months relative to the month of SHS adoption and  $\gamma_\tau$  is the interaction effect of PV capacity at  $\tau$  months. All other variables are defined as they were in the previous three models.

### Potential threats to identification

To address how SHS missclassification error affects our results, we follow the method used in Aigner, 1973 to understand the impact of missclassification on a binary covariate. Although similar to classical attenuation, where the observed variable can be classified as  $Y = Y^* + u$ , binary missclassification introduces additional complexity, so we cannot assume our results are a lower bound without further exploration. Aigner, 1973 shows that the impact of misclassification is state dependent: the probability of error differs when 0 is observed versus 1 being observed. This leads to the use of false positive rates (FPR) and false negative rates (FNR) in binary missclassification analyses. In addition, with binary missclassification, Aigner, 1973 finds that coefficients can be severely distorted when  $\text{FPR} + \text{FNR}$  is high, missclassification is asymmetric, or prevalence of the binary variable is low or high. Because our study does not adhere to strict assumptions including symmetric in missclassification and small FPR/FNR rates, our case will not resemble classical attenuation.

To assess the implication of SHS misclassification, we run a Monte Carlo sensitivity analysis on our

binary SHS indicator. To do this, we first assume  $SHS_{it}$  to be the true binary indicator and  $\widetilde{SHS}_{it}$  to be the observed value.  $\alpha$  is our false positive rate and  $\theta$  is our false negative rate.

$$y_{it} = \delta_1 \text{Outage}_{it} + \delta_2 \widetilde{SHS}_{it} + \delta_3 (\text{Outage}_{it} \times \widetilde{SHS}_{it}) + \eta_i + \tau_m + \tau_y + \varepsilon_{it}, \quad (5)$$

Where,

$$\widetilde{SHS}_{it} = SHS_{it} \cdot (1 - \theta) + (1 - SHS_{it}) \cdot \alpha,$$

and

$$\widetilde{SHS}_{it}^{(n)} = \begin{cases} 1, & \text{with probability } (1 - \theta), & \text{if } SHS_{it} = 1, \\ 0, & \text{with probability } \theta, & \text{if } SHS_{it} = 1, \\ 1, & \text{with probability } \alpha, & \text{if } SHS_{it} = 0, \\ 0, & \text{with probability } (1 - \alpha), & \text{if } SHS_{it} = 0, \end{cases}$$

and  $n$  is the Monte Carlo draw. To classify uncertainty in the PV size, we assess the impact of a model with measurement error:

To calculate the false positive rate (FPR) ( $\alpha$ ) and false negative rate (FNR) ( $\theta$ ), we first take year-specific precision and recall values from instance-based calculations from the model, as well as from post-processing assumptions. We take the M2F FPR and FNR instance-based metrics and reduce error by removing predictions smaller than the size of one panel (1.7 m<sup>2</sup>) and predictions that we can not associate with a building in Cape Town based on spatial intersection or being within 100<sup>2</sup> of a building.

Lastly, because we have a panel of aerial imagery across four years, we use time series assumptions to remove or add panels. We define three assumptions:

1. Rule 1: Delete a single isolated prediction (except 2023)
2. Rule 2: Fill gaps between observed years
3. Rule 3: Forward-fill predictions into 2023 if empty

These rules are based on the assumption that once a PV array is installed on a household, it will not be removed within the four-year time period. In addition, if our model predicts a household has a panel in 2021, for example, but not any other year, then we remove the isolated instance (isolated installations in 2023 are allowed, as it is our last year of imagery). Lastly, any household that still has an SHS after passing through Rule 1, we forward fill to 2023.

Table 4: PV Contracts and SHS Installations by Contract Type and Year

|                                | 2020   | 2021  | 2022   | 2023   |
|--------------------------------|--------|-------|--------|--------|
| <b>M2F</b>                     |        |       |        |        |
| FPR                            | 39%    | 23%   | 23%    | 23%    |
| FNR                            | 32%    | 27%   | 27%    | 27%    |
| Predicted PV                   | 22,376 | 6,893 | 11,976 | 18,142 |
| <b>Time Series Assumptions</b> |        |       |        |        |
| PV added                       | 0      | 3,166 | 1,177  | 1,583  |
| PV removed                     | 15,813 | 1,357 | 1,735  | 0      |
| Total PV                       | 6,563  | 8,702 | 11,418 | 19,725 |
| <b>Monte Carlo</b>             |        |       |        |        |
| FPR                            | 39% *  | 23% * | 18%    | 17%    |
| FNR                            | 32% *  | 2%    | 8%     | 21%    |

Note: \* implies we use the original FPR and FNR rates from the M2F model as calculated values are less than

0 or greater than 100%.

We estimate new FPR and FNR rates based on the number of SHS added and subtracted by the time series assumptions. Note that our values overestimate the model error as we are not able to quantify the improvement in model accuracy after removing small objects or predictions not attached to a household.

## Results

We first evaluate the average impact of SHS adoption on consumption during load shedding through results from Models 1-3 for prepaid and postpaid customers separately. After identifying the average effect, we next look at the impact of SHS ownership over time to understand whether households experience impact for the short or long term. Lastly, we present results from our Monte Carlo analysis to get of sense of how missclassification error from our Mask2Former model may impact our results.

### Impact of Load Shedding and SHS Adoption on Electricity Consumption

We begin by presenting the impact of loadshedding, SHS ownership, and corresponding capacity on monthly electricity consumption behavior for prepaid customers.

Table 5: PanelOLS Estimation Results prepaid

|                                        | (1) Baseline           | (2) SHS Dummy          | (3) SHS Capacity       |
|----------------------------------------|------------------------|------------------------|------------------------|
| Outage duration (hr)                   | -0.0043***<br>(0.0014) | -0.0052<br>(0.0038)    | 0.0011<br>(0.0031)     |
| SHS (binary)                           |                        | -0.2052***<br>(0.0020) |                        |
| SHS capacity (kW)                      |                        |                        | -0.0548***<br>(0.0009) |
| SHS (binary)<br>× Outage duration      |                        | 0.0012<br>(0.0041)     |                        |
| SHS capacity (kW)<br>× Outage duration |                        |                        | -0.0015<br>(0.0009)    |
| <b>Observations</b>                    | 12,736,348             | 12,736,348             | 12,736,348             |
| <b>Entities</b>                        | 414,101                | 414,101                | 414,101                |
| <b>Time Periods</b>                    | 44                     | 44                     | 44                     |

Robust standard errors in parentheses.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1.

Dependent variable: *logged monthly consumption (kwh)*.

Without additional covariates, an hour (or 0.14% of a month) of load shedding corresponds to a 0.43% reduction in monthly consumption, roughly comparable to several hours of average consumption. Households with SHS consume approximately 21% less electricity per month than households without SHS, likely because these systems are wired to use solar generation first with the grid as backup. SHS households do not respond statistically differently to outages than non-SHS households, which reduce consumption by 0.52% per hour of load shedding for non-SHS households.

Including the SHS covariates removes the significant effect of outage duration on monthly electricity consumption for households without an SHS, suggesting the baseline outage effect estimated in model 1 was partly accounting for the SHS effect. Per kilowatt of installed capacity, households with SHS consume approximately

5% less electricity per month than households without SHS. For typical PV systems for prepaid households, this corresponds to 5% for small systems (1kW), 15% for medium systems (3kW) and 40% for large systems (8kW).

Table 6: PanelOLS Estimation Results postpaid

|                                        | (1) Baseline        | (2) SHS Dummy          | (3) SHS Capacity       |
|----------------------------------------|---------------------|------------------------|------------------------|
| Outage duration (hr)                   | -0.0046<br>(0.0035) | -0.0021<br>(0.0297)    | 0.0122**<br>(0.0061)   |
| SHS (binary)                           |                     | -0.2182***<br>(0.0031) |                        |
| SHS capacity (kW)                      |                     |                        | -0.0631***<br>(0.0010) |
| SHS (binary)<br>× Outage duration      |                     | -0.0017<br>(0.0298)    |                        |
| SHS capacity (kW)<br>× Outage duration |                     |                        | -0.0040**<br>(0.0016)  |
| <b>Observations</b>                    | 2,006,056           | 2,006,056              | 2,006,056              |
| <b>Entities</b>                        | 65,308              | 65,308                 | 65,308                 |
| <b>Time Periods</b>                    | 45                  | 45                     | 45                     |

Robust standard errors in parentheses.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1.

Dependent variable: *logged monthly consumption (kwh)*.

Postpaid consumers receiving a household electricity rate exhibit broadly similar patterns to prepaid households. In models 1 and 2, an hour of load shedding does not significantly impact monthly electricity consumption. The SHS ownership indicator in model 2 suggests that households with SHS consume approximately 22% less electricity per month than households without SHS.

Model 3 shows a slight deviation from the other two models. As with prepaid households, the effect of an hour of load shedding on households without SHS becomes positive, suggesting that households increase consumption, though the magnitude is small (1.2%). This positive outage coefficient may reflect load shifting or rebound behavior during outage periods. Households with PV have lower monthly consumption than households without these systems by approximately 6% per kW of PV capacity. They also increase consumption during outages, but by an even more modest amount (0.8%)

## Event Study

We next look at the dynamic impact of SHS adoption and capacity on household electricity consumption using an event-study framework with continuous treatment, following Callaway et al., 2024. Monthly transactions are aggregated into three-month bins to reduce noise and to account for installation date uncertainty. Because aerial imagery identifies the presence of SHS but does not record the exact installation date, we assume that a system first appearing in a given year of imagery was installed six months before the year of imagery. For example, SHS observed in 2022 is assumed to have been installed in June 2021. This timing convention is applied consistently across years, as all imagery was collected in January except for 2021, which was captured in March. We present the results for prepaid and postpaid SHS households below (households without SHS are excluded as they typically have lower average consumption).

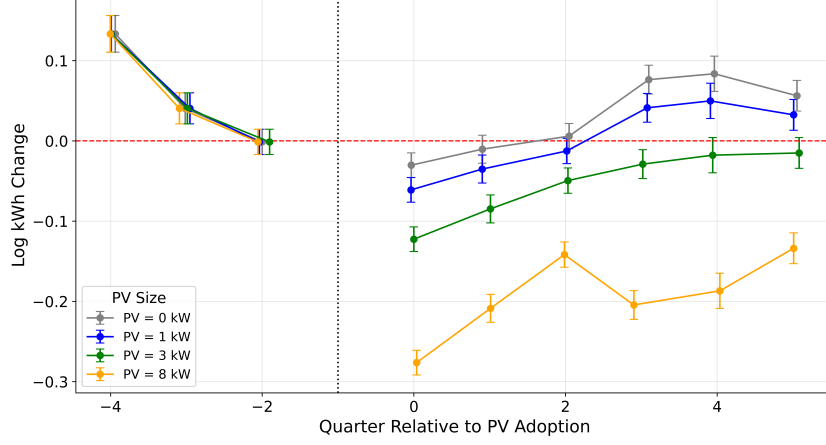


Figure 8: Continuous treatment event study for prepaid households, with the pre-installation period (bin 1: February–April) normalized to zero. Results are shown separately for each SHS capacity quantile, with jitter applied to visually distinguish results by year.

The results presented in Figure 8 indicate that after SHS adoption, households with smaller PV systems ( $\leq 1$  kW) tend to reduce consumption up to 10% right away, consistent with the fact that households initially draw electricity from their PV system, using grid electricity as a secondary source once the PV-generated supply is depleted. However, this effect dissipates over time, with smaller systems leading to increased monthly consumption from the grid compared to baseline approximately 9 months (bin 3) after installation. Because households on the Home User tariff consume roughly 450 kWh/month, or 15 kWh a day, a small system ( $< 1$  kW) can generate up to 5 kWh a day (assuming 5 sunlight hours). At most, this could reduce consumption by about a third. These results suggest behavioral responses among households with SHS, consistent with rebound effects or increased appliance acquisition and usage following reductions in municipal electricity expenditures.

Households with medium to large SHS systems (3 – 8 kW) also reduce consumption after installation (up to 30%), with the effect also lessening over time. Grid consumption for households with medium-sized systems goes back to normal levels a year after adoption (bin 4). Households with larger systems, however, do not appear to go back to normal consumption within the time frame of our analysis (15 months), although the consumption reduction effect does seem to mostly lessen over time.

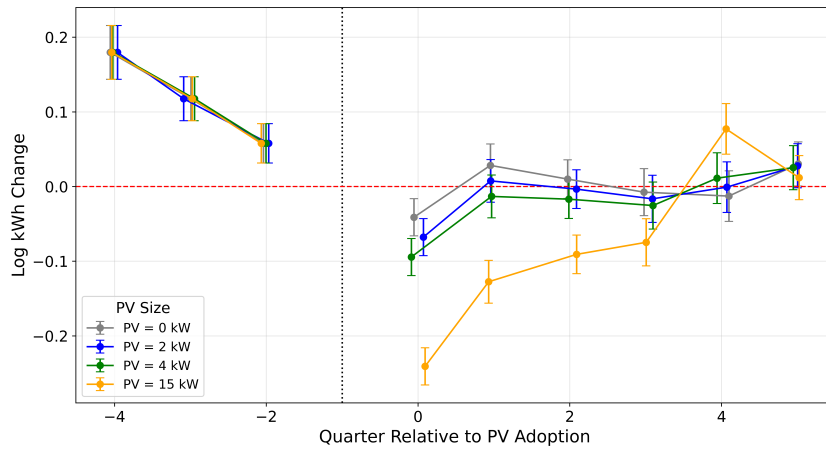


Figure 9: Continuous treatment event study for postpaid households, with the pre-installation period (bin 1: February–April) normalized to zero. Results are shown separately for each SHS capacity quantile, with jitter applied to visually distinguish results by year.

Households with postpaid accounts show a similar behavioral pattern to prepaid households. However, unlike prepaid households, postpaid households return to baseline consumption more quickly, even with large systems. Households that adopt small and medium systems reduce consumption from the grid right away but return to baseline consumption three months later (bin 1) and then do not show evidence of a rebound effect, increased appliance ownership, or usage within the timeframe of this analysis. Postpaid households with large systems (15 kW) behave similarly, but with a larger immediate drop in consumption (around 25%), and returning to baseline (or slightly above) the following year (bin 4).

## Monte Carlo Simulations

We ran a Monte Carlo simulation to understand the impact of SHS identification uncertainty in our dataset. We conduct 500 simulations for our first model that included the SHS dummy variable.

Table 7: Monte Carlo simulation summary for the effect of SHS adoption on electricity consumption

|                    | Postpaid | Prepaid |
|--------------------|----------|---------|
| Mean Coefficient   | −0.008   |         |
| Standard Deviation | 0.001    |         |
| Bias               | 0.213    |         |
| RMSE               | 0.213    |         |
| Coverage 95% CI    | 0        |         |

The results suggest that PV assignment misclassification can almost entirely mask the true reduction in consumption from the grid for SHS households. Although the estimated coefficients are precise due to low standard deviation, the error is dominated by systematic (bias), rather than random (variance) error, as estimates are far from the baseline value. In addition, the simulated 95% confidence intervals do not capture the baseline estimate. This suggests that the approximately 20% decrease in monthly consumption for prepaid and postpaid households with SHS is likely an underestimate.

## Additional Robustness Checks

We next run our three regression models for both prepaid and postpaid households using only households with registered PV in 2021 and 2022, where all SHS adoption was manually verified.

Results soon unless you think its too much

## Discussion

### Impact of Load Shedding on Grid Consumption

The results indicate that loadshedding does not reduce overall electricity consumption for either prepaid or postpaid households. This suggests that households shift usage to before or after a load-shedding period and may also experience a rebound effect. In Cape Town, a typical load shedding period can last for two to two and a half hours, a short enough window that most flexible electricity usage can be rescheduled rather than lost. For instance, a consumer intending to run a washing machine during an outage may instead operate it beforehand. Most household loads are flexible in this way, with the exception of continuously operating appliances, such as refrigerators, whose consumption cannot be rescheduled.

When a consumer adopts an SHS, their marginal cost of electricity initially decreases, due to the immediate dip in consumption right after installation. A household is wired to consume electricity generated from the SHS first and then resort to the grid for any unmet demand. Because SHS systems only incur upfront installation costs and no per-unit generation costs, households experience an immediate decrease in the marginal cost of electricity upon adoption. This allows households to accommodate rebound effects with a smaller impact on their monthly electricity expenditures. It may also allow households to invest in additional appliances. These rebound behaviors offer an explanation for the event study results, showing most households resume or exceed pre-SHS adoption consumption levels.

On average, households with SHS consumption roughly 5% less electricity per kW of installed capacity compared to households without SHS. Postpaid households are also marginally affected by outages: each hour of load shedding increases consumption by approximately 1% for households without PV, and by 0.8% per kW of SHS capacity for SHS households. The impact of outages on SHS households remains modest for average system sizes. This modest reduction in consumption during outages likely reflects that households attempt to maintain normal usage, but their SHS is not large enough to fully sustain it.

## Impact of SHS on Grid Consumption

To understand the impact of SHS adoption on municipality revenue, we calculate the loss in sales assuming SHS households consume 20% less electricity than an average household in the same tariff group. We take yearly tariff values to estimate the loss of revenue for each tariff category.

Table 8: Revenue loss for each tariff group

|                                  | 2020                  | 2021                  | 2022                   | 2023                  |
|----------------------------------|-----------------------|-----------------------|------------------------|-----------------------|
| Total Residential Sales          | 183,000,000           | 191,000,000           | 187,000,000            | 168,000,000           |
| <b>SHS Prepaid Revenue Loss</b>  |                       |                       |                        |                       |
| LifeLine                         | 1,188                 | 1,463                 | 2,480                  | 3,366                 |
| Domestic                         | 6,831                 | 10,458                | 14,078                 | 21,357                |
| Home User                        | 32,421                | 46,742                | 87,669                 | 176,918               |
| <b>SHS Postpaid Revenue Loss</b> |                       |                       |                        |                       |
| Home User                        | 31,360                | 40,026                | 58,212                 | 90,235                |
| <b>Total</b>                     | <b>71,800 (0.04%)</b> | <b>98,689 (0.05%)</b> | <b>162,439 (0.09%)</b> | <b>291,876 (0.2%)</b> |

*Note: this assumes SHS households stay within Block 1 of their tariff group.*

Loss in residential sales is mainly due to prepaid and postpaid households on the Home User tariff. Losses from LifeLine and Domestic tariff groups have had almost a negligible impact on electricity sales revenue from 2020 – 2023. In fact, achieving the same revenue loss as that generated by prepaid Home User households would require SHS adoption by roughly one in ten Lifeline households in 2020 and nearly one in three in 2023. By comparison, prepaid Home User households accounted for only 2.5% of all Home User customers in 2020 and 7.2% in 2023. This suggests that encouraging SHS adoption among LifeLine households is unlikely to substantially reduce municipal electricity revenue. A low-income SHS subsidy would also spread load shedding resilience to lower-income or lower electricity consumption households, a benefit that is currently concentrated among higher-consumption households.

However, encouraging SHS adoption across all households in Cape Town does not seem to be as detrimental as suggested in other works (XX), and could potentially help the utility and municipality reduce load shedding

on the grid as household demand decreases. Future work should be conducted on these benefits as Cape Town works to eliminate load shedding and decrease fossil-fuel emissions from electricity-generating resources.

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## Appendix

### A Masked-attention Mask Transformer (Mask2Former)

Our study uses the Masked-attention Mask Transformer (Mask2Former) model (Cheng et al., 2022), building off the MaskFormer model (Cheng et al., 2021) to identify SHS in aerial imagery of Cape Town. It comprises of a Swin-Small backbone that performs feature extraction, pixel decoder which transforms the low resolution backbone features to high resolution, and transformer decoder that generates the predictions. Mask2Former, a recently developed model, has been used to identify cancer cells in pathology images (Sheng et al., 2023), modeling lettuce seedlings (Wei et al., 2024), remote sensing applications (Qiao et al., 2023), and precision agriculture (Darbyshire et al., 2023).

Our aerial imagery dataset covers the entirety of Cape Town (see Figure A10), and each image (or grid cell patch) has a resolution of 8cm per pixel. Further details can be seen in Figure A10 and Table A9

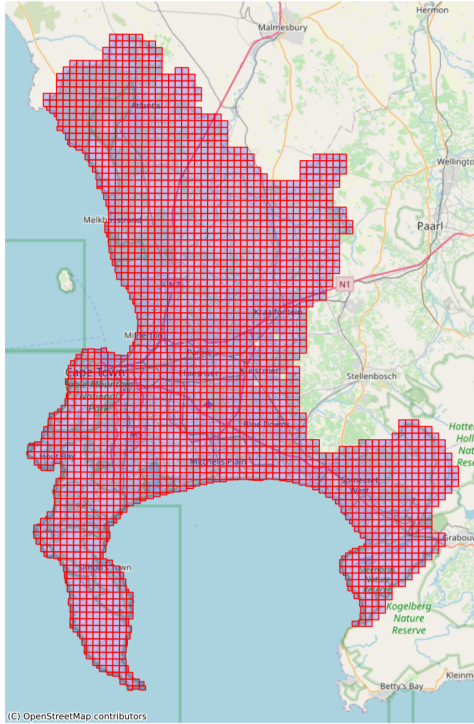


Table A9: Summary of aerial imagery dataset by year.

| Year | Number of images | Total GB |
|------|------------------|----------|
| 2023 | 2802             | 511      |
| 2022 | 2819             | 663      |
| 2021 | 2805             | 686      |

Figure A10: Coverage of 2023 aerial imagery for Cape Town.

To create our training set, we used QGIS to hand-label 2,033 solar panels, 1,983 solar water heaters, and 1,146 pool heaters in 15 images (0.5% of the 2023 aerial imagery) from a stratified sample of 2023 images. Each annotation was then checked by another annotator to confirm the accuracy of the boundaries and the assigned label through a custom annotation validation tool. The second annotator can then confirm the annotation label accuracy, recategorize it, or redraw it. Figure A14 is an example from our annotated dataset, where we classify solar objects into either PV panels (or, SHS system), water heaters, or pool heaters.



Figure A11: Annotations (translucent, and filled in) with PV panel in yellow, PV water heater in red, and PV pool heater in blue

To prepare this data for modeling, we tile each image and all corresponding masks into  $320 \times 320$  pixels to fit with the model specifications.



Figure A12: Example tile

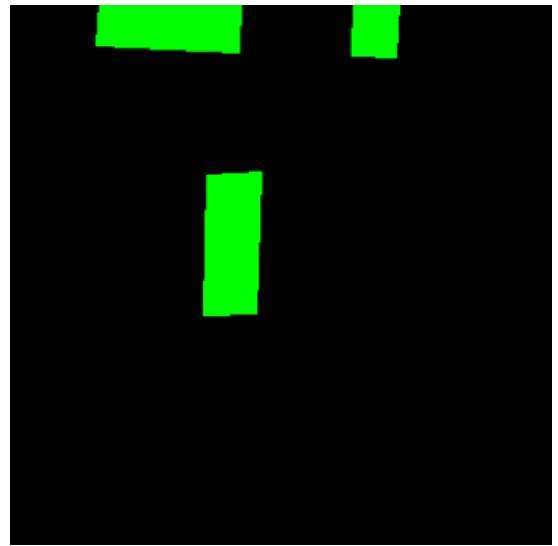


Figure A13: Example mask

We use 0.5% of the aerial imagery from 2023 to train our model, which we then apply to years 2019-2023 to make predictions for the locations and sizes of all solar home systems in the city. We train the model with 60 epochs and 0.00035 learning rate on an 8 GPU machine.

## Results



Figure A14: Annotations (translucent, and filled in) and prediction (outline) with PV panel in yellow, PV water heater in red, and PV pool heater in blue

We first use pixel-level metrics to assess the model accuracy. Results are shown in Table A10.

Table A10: Pixel-based evaluation metrics for SHS detection, IoU = Intersection over Union

| Class           | IoU    | Precision | Recall | F1     | Support |
|-----------------|--------|-----------|--------|--------|---------|
| PV panels       | 0.7232 | 0.7932    | 0.8911 | 0.8393 | 310,577 |
| PV water heater | 0.6084 | 0.8194    | 0.7027 | 0.7566 | 76,301  |
| PV pool heater  | 0.7507 | 0.8650    | 0.8504 | 0.8576 | 251,887 |

## Post Processing

After connecting adjacent pixels, forming polygons, and then combining any neighboring polygons, we can compute instance-based evaluation metrics with an intersection over union threshold of 0.5.

Table A11: Instance-based evaluation metrics for SHS detection

| Class           | TP  | FP  | FN  | Precision | Recall |
|-----------------|-----|-----|-----|-----------|--------|
| PV panels       | 245 | 329 | 95  | 0.4268    | 0.7205 |
| PV water heater | 191 | 301 | 102 | 0.3882    | 0.6518 |
| PV pool heater  | 163 | 162 | 81  | 0.5015    | 0.6680 |

Our evaluation metrics show a substantial amount of false positives. To improve accuracy, we impose two restrictions on the dataset. First, we remove any SHS predictions with an area  $\leq 1.7 \text{ m}^2$ , the typical size of a residential solar panel (EcoFlow, 2025). Next, we take the centroid of each solar panel and overlay it onto building polygons (Overture Maps Foundation, 2025), assigning each SHS to a building for each year. We allow a building to have more than one SHS system.



Table A12: Filtering of prediction data

|                          | <b>2021</b> | <b>2022</b> | <b>2023</b> |
|--------------------------|-------------|-------------|-------------|
| Total predictions        | 150,591     | 211,770     | 153,353     |
| SHS with area $\leq 1.7$ | 74,001      | 109,753     | 61,353      |
| SHS not on building      | 82,342      | 130,850     | 50,257      |
| Remaining predictions    | 41,013      | 50,065      | 71,257      |

## B SHS Capacity Calculation

To derive capacity estimates from our surface area predictions, we perform the following calculation:

$$\text{Capacity (W)} = \text{Area of PV array (m}^2\text{)} \times \frac{\text{Average capacity per panel (W)}}{\text{Average panel size (m}^2\text{)}}$$

Where we assume an average panel size of  $1.7\text{m}^2$  and wattage of 400 W per panel (EcoFlow, 2025).